Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/08/2016
PI Name:	Judex, Stefan Ph.D.		
Project Title:	Genes that Predict the Loss of Bone during Weightlessness		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
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No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 10/01/2017 (9/30/17) per NSSC information (Ed., 9/8/16) NOTE: Extended to 10/31/2016 per NSSC information (Ed., 9/10/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Leah Rae (Jackson Laboratory)		
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Task Description:	The amount of bone loss between individual astronauts is highly variable with some astronauts losing large amounts of bone while others are largely unaffected. This variability is accounted for, at least to a large extent, by genetic variations between individuals. Unfortunately, the specific genes that modulate this genetic sensitivity of bone loss in microgravity are unknown. To this end, we have recently completed a genetic study in which we identified specific regions on chromosomes (QTL or Quantitative Trait Loci) that either add to or reduce the propensity of an individual to lose bone when normal weightbearing is removed from the musculoskeleton. While providing critical information, these QTL still contain hundreds of genes, rather than a single one. Here, we propose first to greatly reduce the number of candidate genes with bioinformatics tools and then experimentally test the shortlist of candidate genes using knockout mice available from the Jackson Laboratory's (JAX) Genetic Resource Repository and the JAX Knock Out Mouse Project (KOMP) production center. The identification of single genes modulating bone loss during weightlessness can be used as a critical diagnostic tool for the identification of astronauts most in need of countermeasures as well as to unravel novel drug targets that preserve bone health in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal directly addresses one of the six highest priority recommendations made in the National Research Council's 2011 Decadal Survey Report, "Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era" (http://www.nap.edu/), namely recommendation AH3: "Bone loss studies of genetically altered mice exposed to weightlessness are strongly recommended." Defining some of the specific genes that cause differences in the skeletal sensitivity to weightlessness will not only provide novel diagnostic and therapeutic means of combating bone loss in space but also on Earth in individuals subjected to bedrest or suffering from paralysis.
Task Progress:	Year Two and Year Three progress reports discussed microcomputed tomography indices from CD44, FGF2, ESR1, ESR2, FAS, and NOD2 genes. Here, we report our results from Pak 6 (p21 activated kinase 6) mice. We also report osteoblastic and osteoclastic activity in ESR1, ESR2, and Pak6 null mice. Physiological levels of estrogen are critical for maintaining bone health. This is evident from extensive bone deterioration observed in women with low circulating estrogen post menopause or after surgical removal of ovaries [1, 2]. Estrogen is a steroidal hormone that exerts most of its bone specific actions via its interaction with its receptors (estrogen receptors, ESR; ESR1 and ESR2) [3]. We previously reported that ESR1-/- mice had greater trabecular and cortical microcomputed tomography indices at baseline and were protected from the catabolic effects of hindlimb unloading (HLU). ESR2 -/- mice had greater trabecular and cortical indices at baseline and were more susceptible to hindlimb unloading losses than wildtype (WT) C57BL/6J mice. Here we report that the resistance to HLU in ESR1 -/- mice were due to lower suppression of osteoblastic number and surface during unloading, while ESR2 -/- mice were more susceptible to losses because of greater suppression of osteoblastic number and surface during unloading when compared to their WT counterparts. Pak6 (p21 activated kinase 6) gene has been known to interact with ESR1 and suppress the transcriptional activity of ESR1 [4, 5]. Therefore, we hypothesized that Pak6 null mice will exhibit lower susceptibility to unloading similar to the response exhibited by ESR1 -/- mice. To test the function of ESR1, ESR2, and Pak6 genes during mechanical unloading, we divided ESR1 KO (ESR1-/-, n=36), ESR2 KO (ESR2 -/-, n=20), and Pak6 KO (n=20) into baseline control (BC), age-matched control (AC), and hindlimb unloaded (HLU) groups. All KO mice were backcrossed to C57BL/6J (B6J) mice for at least 8 generations. Therefore, B6J mice were used as wildtype (WT) controls. At the end of the experiment, the mice were sacrificed and femurs were harvested and fixed in 10% neutral buffered formalin and then stored in 70% ethanol for ex vivo microcomputed tomography scanning (µCT 40, Scanco Medical, SUI) and histology. Ex vivo µCT scanning was performed on the distal femoral metaphysis to quantify trabecular and cortical microarchitecture at a 10 µm resolution. After microcomputed tomography scanning, femora were decalcified in 14% EDTA (ethylene diamine tetra acetic acid) and the bones were embedded in paraffin, and at least one 5µm section per bone was stained with either toluidine blue or tartrate resistant acid phosphatase (TRAP). In a region similar to the one analyzed with µCT, osteoblast surface per bone surface (Ob.S/BS, %) and osteoblast number per bone surface (Ob.N/BS) were quantified using toluidine blue stain, while TRAP stain quantified osteoclast surface to bone surface (Oc.S/BS %). In cortical bone, these osteoblastic and osteoclastic indices were quantified and normalized separately either to the endocortical or periosteal perimeter (Ec.Pm/Pe.Pm). Ob.S/Ec.Pm, Ob.N/ Ec.Pm, Oc. S/ Ec.Pm, Ob.S/Pe. Pm, Ob.N/ Pe.Pm, and Oc. S/ Pe.Pm were measured using Osteomeasure (Osteometrics, Inc., Decatur, GA, USA). The effect of gene deletion was tested by comparing age matched control KO mice to WT mice using two-tailed, independent samples T tests. The impact of HLU within a given genetic strain was tested for significance via one-way ANOVA and Tukey post-hoc test. Differences in magnitude of bone deterioration between WT and KO mice were tested by two-way ANOVA. We had previously reported that 11 wk old age-matched ESR1-/- mice and ESR2-/- mice had greater trabecular and cortical µCT indices than WT control. Further, we also reported that trabecular bone of HLU ESR1-/- mice were resistant to the catabolic effects with no differences in bone volume fraction and minimal responses in trabecular thickness (Tb.Th) while trabecular bone of ESR2 -/- were extremely susceptible to HLU with significantly greater losses than WT mice. Here we report that Pak6 -/- mice during normal cage ambulation, similar to ESR1 and ESR2 -/- mice had 31%, 7%, 9%, and 9% greater connectivity density (Conn.D), trabecular number (Tb.N), cortical area (Ct.Ar), and cortical thickness (Ct.Th) when compared to WT AC mice (all $P<0.05$). HLU WT mice had 31% lower BV/TV, 5% lower Tb.N, and 19% lower Tb.Th while HLU Pak6-/- mice had 32% and 18% lower BV/TV and Tb.th when compared to corresponding AC mice (all $P<0.05$). HLU Pak6 -/- mice had 16% and 10 % lower Ct.Ar and Ct.Th, while WT mice had 15% lower Ct.Ar and 7% lower Ct.Th when compared to corresponding AC mice. However, the HLU response of the Pak6 KO mice was statistically not different from WT mice in both trabecular and cortical compartments. ESR1, ESR2, and Pak6 null mice had lower bone resorptive TRAP activity (Oc.Pm/B.Pm) in trabecular bone. However, ESR1 -/- (3%) and Pak6 -/- (9%) mice had greater osteoclastic activity than WT mice (2%) on the endocortical surface of the cortical bone (all $P<0.05$). Periosteal TRAP activity on the other hand, was lower in ESR1-/- (0.6%) when compared to WT (1.6%) mice ($P<0.05$). Pak6 -/- mice had 22%, and 40% lower osteoblastic counts on the trabecular and on the periosteal cortical surface ($P<0.05$). In ESR2-/- mice, osteoblastic activity on trabecular, endocortical, and periosteal surfaces were not different from WT mice. These data suggest that under physiological conditions, the greater trabecular and cortical indices in ESR1-/-, ESR2 -/-, and Pak6 -/- mice were primarily due to lower osteoclastic activity. The differences in HLU trabecular response of ESR2 -/- mice were due to greater suppression of Ob.N/BS (56% vs 30%) and Ob.S/BS (58% vs 40%) when comparing unloaded mice to AC groups in ESR2 -/- mice and WT mice (Gene*HLU, $P<0.05$). Resistance to HLU observed in ESR1 -/- mice was due to lack of suppression of osteoblastic

indices in this KO strain (nsd) when subjected to hindlimb unloading. HLU Pak6 -/- mice had 47% and 50% lower Ob.N/BS and Ob.S/BS when compared to corresponding AC mice (all P<0.05) but the response was not different from WT mice. These results suggest that the trabecular differences in HLU responses between ESR1 -/-, ESR2 -/-, and WT mice are mediated largely by differences in osteoblastic activity rather than osteoclastic activity. In cortical bone, HLU did not affect osteoblastic number and activity on the endocortical or periosteal surface of KO mice. In WT mice, Ob.N, and Ob.S were 30% and 41% lower in HLU when compared to corresponding AC mice (all P<0.05), while no differences were observed on the periosteal surface. No differences in osteoclastic activity was observed on the endocortical surface of both KO and WT mice. On the periosteal surface, no differences in osteoclastic activity were detected between HLU and AC ESR1-/- mice but HLU WT B6J mice had 300% greater osteoclastic activity than AC mice (Gene*HLU, P<0.05). Osteoclastic activity of HLU ESR2 -/- and Pak6 -/- mice was not different from WT mice on the periosteal surface.

Taken together, these results highlight the importance of ESR1, ESR2, and Pak6 genes in skeletal function during physiological and hindlimb unloading conditions. Further, the results suggest that the greater susceptibility and resistance to HLU in ESR2 -/- and ESR1 -/- mice was largely due to greater/lower suppression of osteoblastic indices rather than changes in osteoclastic activity.

References

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